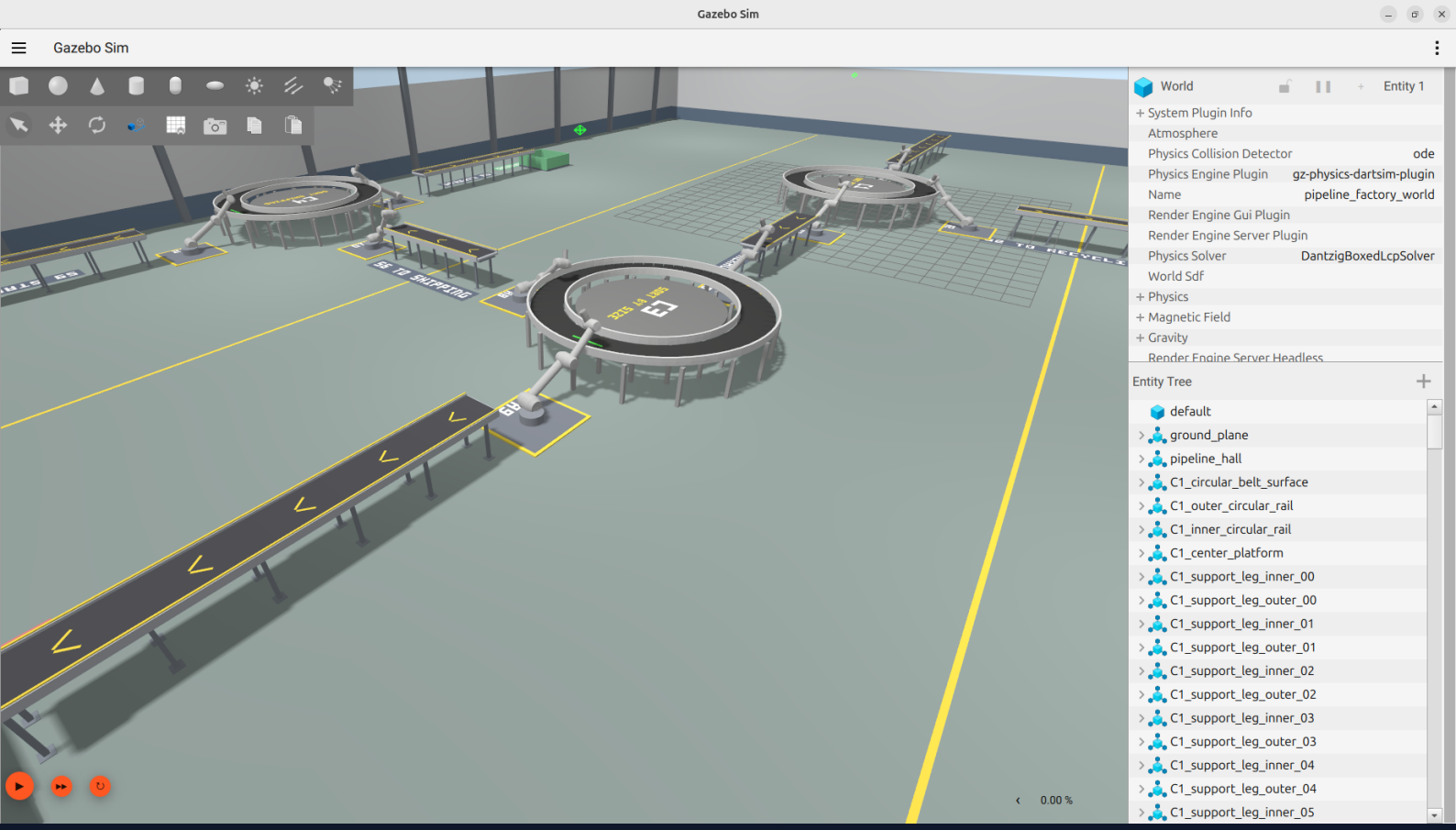


REPORT 02 · FACTORY BENT SIMULATION WITH A BCR ROBOT ARM

Evolution Timeline

From one straight conveyor (March 2026) to a
12-arm factory with a vision lab (November 2026)

ROS 2 Jazzy · Gazebo Harmonic · OpenCV · PyQt5 · BCR 7-axis arm

Hüseyin Tozluyurt · November 2026

How to read this timeline

Before each significant change, a backup of the files it touched was saved with a suffix such as `.bak_before_<phase>` (for example `COMMANDS.md.bak_before_sidecam` or `vision_pick_place.py.bak_before_bins`). The modification time of each backup marks the moment a phase began. Comparing consecutive backups shows what that phase added. This report rebuilds the history from those snapshots, the git log, and the project's work threads.

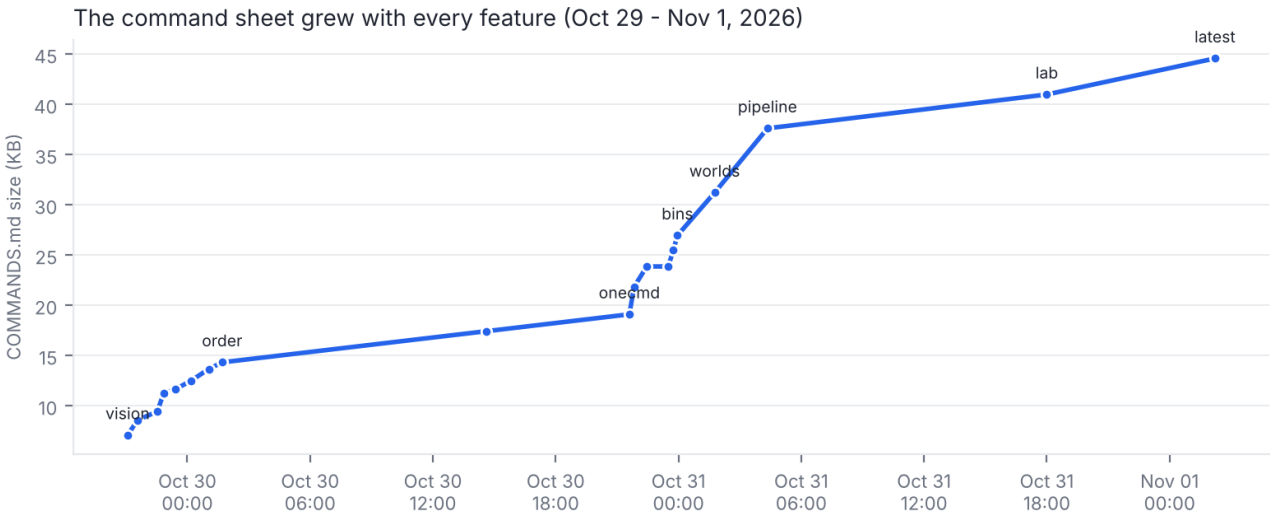


Figure 1. The size of COMMANDS.md at each backup. Steep parts are the busiest periods: Oct 29 evening, Oct 30 night and the early hours of Oct 31.

Phase by phase

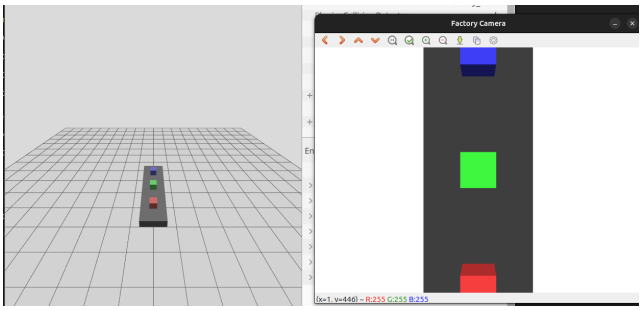
When	Phase	What changed and why
Mar 16–17, 2026	Beginning	Repository created. A straight conveyor world (`factory_conveyor.sdf`), an overhead camera bridged to ROS, `camera_viewer.py`, and the first detection scripts (threshold + contours, then tracking and counting with measurements logged to a txt file).
Apr – Oct 2026	Arm cell	Circular and dual circular worlds, BCR arm integration (`factory_with_bcr_arm.launch.py`), a PyQt5 arm UI with preset poses, and an endpoint attacher that fakes the grasp with `set_pose`. Pick and place still worked only by hand.
Oct 26	Exact attach	The attacher was rewritten (V2.9). The product's top face snaps to `gripper_tip`, it follows the arm through the gz-transport bindings (no CLI lag), and it is released from the exact pose it was carried at. Package dependencies and the broken fake-gripper entry were fixed.
Oct 29 20:43	COMMANDS.md	One command sheet for building, launching every world, and the attacher, UI, detection and debugging commands.
Oct 29 21:05	vision	**Vision pick and place**. `ring_object_detector` (grayscale, Canny, contours, pixel to world) and `vision_pick_place` (URDF IK, pick still objects in a zone, place on ring 2).
Oct 29 21:34	station	**Catching moving products**. The arm hovers over a catch station. A ROI tracker predicts when a product enters the circle, accounting for the catch latency.
Oct 29 22:32	sidecam	**Side camera cross-check**. A second camera at belt height must confirm the product. The product is then held on the belt, and the arm descends onto its measured top.
Oct 29 22:51	carry	Lift after the grab, capped at the arm's reach, then one swing to ring 2, with a short settle before the release.
Oct 29 23:24	bentstop	The whole bent 1 stops while the product is held, and starts again once the arm has it. The lift uses the shoulder joint only. Exact place slots on ring 2.
Oct 30 00:11	ring2	Bent 2 moves too. The **release-zone camera** waits for an empty zone, and bent 2 is stopped during the release. Separate stop and start services for each bent.
Oct 30 01:03	home	The arm waits at home instead of hovering. **Six movement styles** (`standard`, `mirror`, `high_carry`, `low_fast`, `full_turn`, `reverse_return`), chosen by the object's height.
Oct 30 01:42	order	**Product Order GUI** (PyQt5) and the `/spawn_order` topic. 16 product types, spawned an equal arc distance apart on bent 1.
Oct 30 14:36	pictures	30 product types including spheres. A **photo camera** in the world and `capture_product_images` provide a picture grid in the GUI.
Oct 30 21:36	onecmd	`run_factory.sh`: the order window first, then `factory_full.launch.py` starts everything. A .desktop launcher.
Oct 30 21:50	speed	Bent 2 orders, **one belt speed** on both bents (bent 2 had run 11 % faster), simulation speed (real-time factor) and a run-time limit.
Oct 30 22:27	gripper / style	A **dark factory theme** for the GUI (status light, search, drag-to-sort). **World decor**: hall, fence, stack lights, racks, desk, signs. A plain world is kept for slow PCs.
Oct 30 23:45	grasp	**Close grasp**: a two-step descent, and the grip happens only when FK says the tip is within 1.5 cm of the contact point. Movement style selectable in the GUI.
Oct 30 23:57	bins	**Trash and recycling bins** with collisions. A "Goes to" choice for each product type, and auto-sort by material.

When	Phase	What changed and why
Oct 31 00:08–01:01	pin, binfuncs, bindrop, binclear, fall	The held product is pinned while the gripper descends. `put_in_trash()` / `drop_in_recycle_bin()`. A high travel over the bin centre. IK postures that hit a bin wall are rejected. The free fall into the bin is simulated.
Oct 31 01:46	worlds	The **single circular** and **straight** worlds were rebuilt around the same cell. `WORLD_LAYOUTS` tells every node the belt geometry. On the straight belt, products fall into a collection box.
Oct 31 03:34–03:52	boxdrop, exp	Free fall into the collection box. **Three experimental worlds** that mix straight and circular bents. One launcher per world.
Oct 31 04:21	pipeline	**Pipeline factory**: 4 sorting rings, 9 lines, 12 arms, 5 docks, generated from `pipeline_layout.py` (layout only).
Oct 31 17:32	hangfix	Products hung in the air past the belt end because the belt-end check crashed the attacher. It now reads the pose stream and logs errors instead.
Oct 31 18:00	lab	The **vision lab**: OpenCV/YOLO detectors plugged into the controller in every world, `main(lab)` scripts, lab files, and offline scoring.
Oct 31 19:00–20:00	pipeline lab	A pipeline lab world (a camera and bins at each arm, movable arm links), whole-factory sorting, and a demo in three acts.
Nov 1 00:00–01:00	test, realistic, custom	The straight conveyor was tested on the real PC (products spawn and move). **100 realistic products**. **Custom products** from STL/OBJ/DAE files.
Nov 1 05:19	pipeline tests	8 pipeline test cases with PASS/FAIL for each product along chains of up to 6 arms.

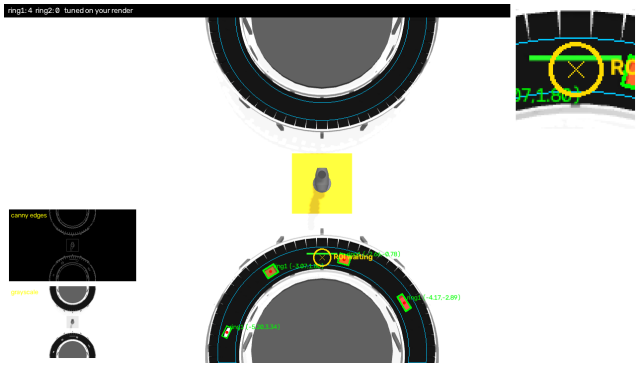
Five eras

1. Simulation scaffold (March to October 2026)

This era built the worlds, cameras and arm. Detection was a standalone script with no link to ROS coordinates, and the arm was moved by hand.



Mar 2026: the straight conveyor and camera view



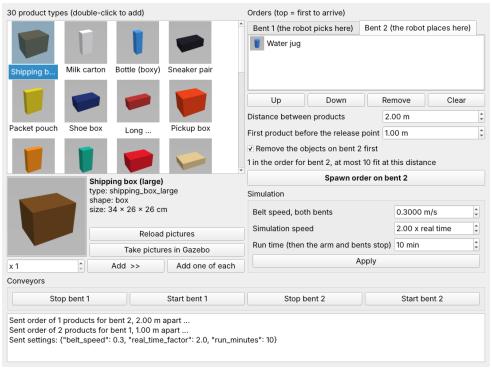
Oct 29: tuned ring detection

2. Closing the loop (Oct 29)

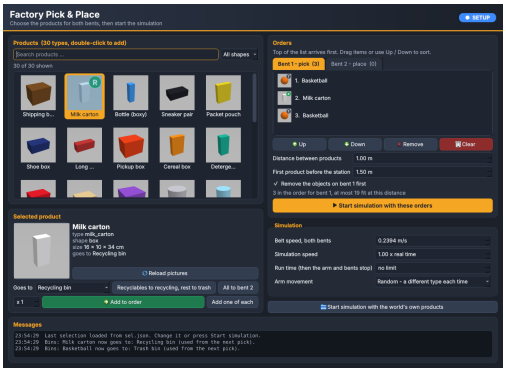
In one evening the project went from detection only to a camera-driven arm. The arm first picked still objects, then caught moving products, then cross-checked with a second camera and stopped the bent while it grasped.

3. Usability (Oct 30)

Once the loop worked, the focus moved to the operator. This brought the order GUI, product pictures, one-command start, speed control, the dark theme and factory decor, and finally bins as a second kind of destination.



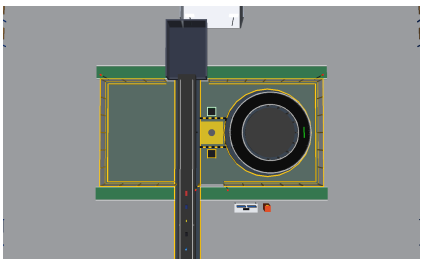
Order GUI, first version



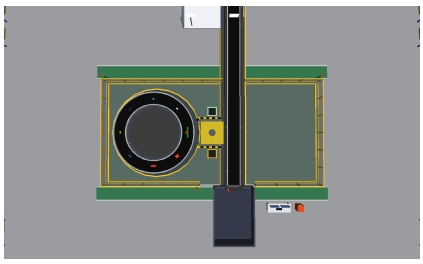
Order GUI, dark theme with bins

4. Generalisation (Oct 31 night)

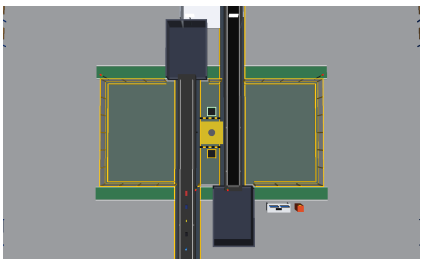
The cell was rebuilt so that any bent shape works. This gave six cell worlds from one layout table, and then a whole factory of 12 cells.



straight → circular



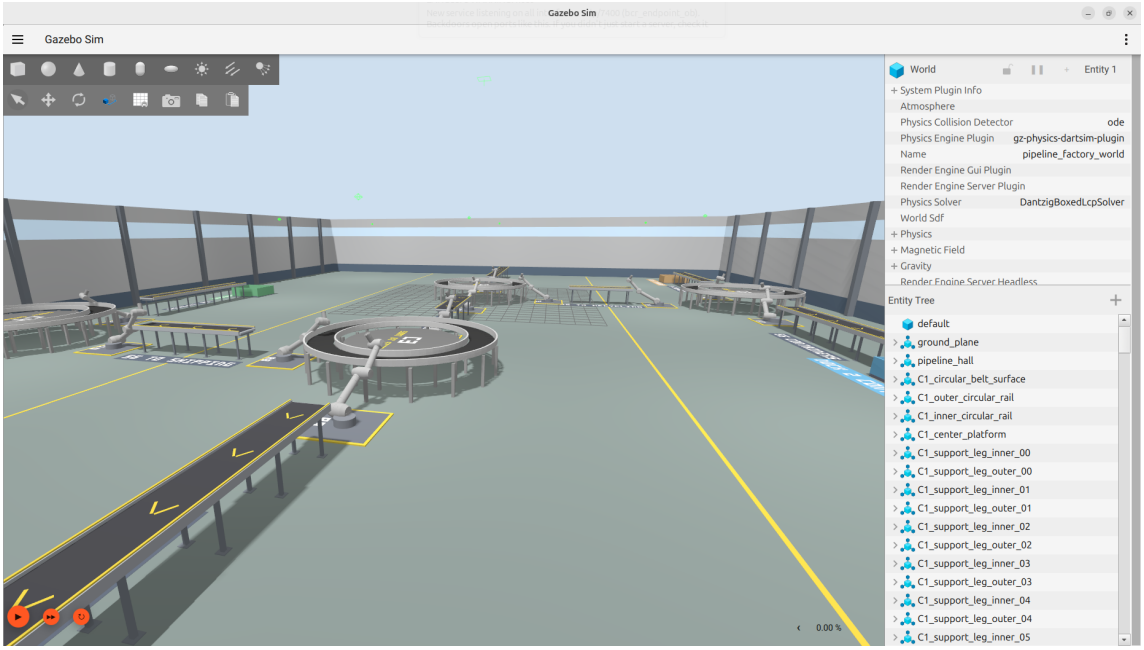
circular → straight



straight → straight

5. Research platform (Oct 31 to Nov 1)

The fixed detector became a slot. The vision lab lets anyone plug in an OpenCV or YOLO detector, run it in any world, score it offline, and test multi-arm routes. The products were made realistic, and users can add their own.



The pipeline factory hall (Oct 31, 07:24).

Backups left behind

Suffix	Files
`.bak_before_vision` … `.bak_before_lab`	18 snapshots of COMMANDS.md
`.bak_before_pin/bindrop/fall/worlds/boxdrop/exp/hangfix`	bcr_endpoint_object_attacher.py (7 versions)
`.bak_before_binfuncs/bindrop/binclear/worlds`	vision_pick_place.py (4 versions)
`.bak_before_worlds/exp`	product_catalog.py, ring_object_detector.py, order_gui.py, run_factory.sh
`.bak_before_photocam`	dual_circular_factory_conveyor.sdf
`.bak_before_order/pictures/ring2/sidecam/vision/lab/custom/realistic`	setup.py (entry points grew phase by phase)

Each phase was written to keep earlier commands working. New behaviour was added behind launch arguments with the old behaviour as an option (for example `mode:=pick_zone`, `use_side_camera:=false`, `wait_at:=station`, `run_detector:=false`), so every step in this timeline can still be run today.